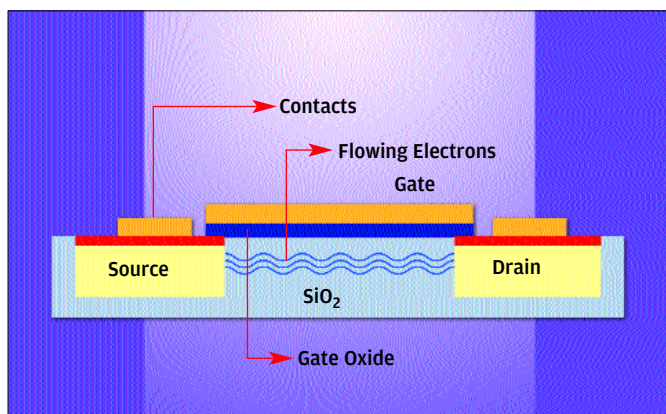




The (simple enough) structure of a typical MOSFET. The electrons flow from the source to the drain through silicon, and are controlled via a voltage applied to the Gate

mesh rolled up, as in the figure alongside (See figure A *typical single-walled carbon nanotube*). It is these structures we're interested in; we'll refer to them simply as "nanotubes"—and we'll come back to them later.

The All-Empowering Transistor

To tie up some of the things we've talked about, the idea is that the transistor is the basic building block of most ICs (Integrated Circuits), and it is transistors that do the actual, atomic-level switching of ones and zeroes. When you say a computer works in binary, we mean that it handles only ones and zeroes—simple enough; but this is true even at a much more fundamental, physical level. All of computing is, in a hardware sense, about switching between a high voltage and a low voltage. It's the transistors that do that.

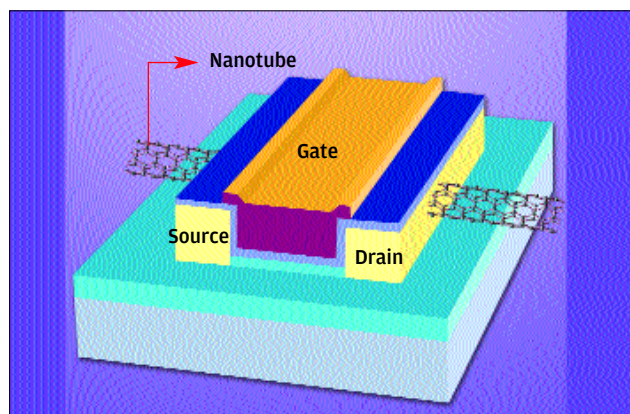
Now, when we talk about the 65-nm manufacturing process, we mean that the conducting lines on the chip's silicon wafer that connect the transistor are 65 nanometres wide. As this figure goes down—that is, as we move to the 40nm or even 20nm process—the transistors are getting smaller. As they get smaller, the chips they comprise get faster, more powerful, and all that. You know the rest of the story—the world demands more computing power, there's a lot of money to be made, and so on—hence we keep shrinking the transistors, amongst other components.

Around 2020, it will not be possible to shrink current-gen-type silicon transistors any more, and that's where nanotubes come in. They will replace the silicon, co-existing with silicon in the interim, and Dr Moore will live on in our hearts and minds.

But to understand what's going to happen in 2020, we need to understand how a transistor works. A little digression...

The Three-Legged Beast...

...called the transistor revolutionised electronics, and therefore everything that depended on it, when it was invented in 1947 by Shockley, Brattain and Bardeen. It's a box with three legs—the source, drain, and gate. The source is where electrons come from; the drain is where



This is something like a MOSFET that has a nanotube built into it. Electrons flow through the nanotube instead of through silicon

they go out from—and the path through which they pass is called the channel. The gate is what controls the flow: an appropriate voltage applied to it reduces the conducting ability of the silicon channel inside, and there won't be any flow of electrons—gate locked. The gate can be opened by a different voltage applied to it. There needs to be an isolation between the channel and the gate, which in the case of MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors—one type of a transistor that's used most frequently in ICs today) is a layer of insulating silicon dioxide.

Applying different voltages to the gate therefore has the effect of the transistor storing either a one or a zero—that is, a bit. The transistor is thus a switching device. In a nutshell, that's how you get your ones and zeroes—voltages applied to the gates of transistors.

The Shrinkage Barrier

For decades now, transistors have kept getting smaller. So what'll happen in 2020? Not to get into the details, MOSFETs are the most common type, like we said. Taking the MOSFET as an example, several things will happen when you try to shrink it beyond a certain point—to be precise, when you decrease the channel length and the thickness of the gate insulator.

The quantum mechanical effect called tunnelling (see *Jargon Buster*), amongst other effects, begins to take over. The leakage currents that result are too large, and render the transistor possibly useless as a switch. On top of that, leakage currents add to the power consumption of devices scaled at this level, as if the current high power consumption weren't enough. Then there's the matter of the thickness of the metal wires that connect the transistor to the rest of the circuit: this needs to be reduced, too. And reducing this width leads, by the laws, to increased resistance—meaning a slowing down of performance, and the phenomenon called electromigration (see *Jargon Buster*). Electromigration decreases the reliability of the IC, and can even sometimes lead to the loss of a connection—and hence, failure of the entire circuit.